

A proof of Zeilberger’s recurrence for solid standard Young tableaux of shape $[[n, n], [n, 1]]$

Jaideep Sai Padhi*

August 2026

Keywords: solid standard Young tableaux; Kreweras walks; algebraic generating functions; kernel method; P-recursive sequences.

MSC 2020: 05A15 (primary); 05A19, 05E10 (secondary).

Abstract

Let $g(n)$ denote the number of solid standard Young tableaux of the two-layer shape $[[n, n], [n, 1]]$. In his *First Rigorous Challenge* accompanying the solid-SYT project, Zeilberger observed empirically that $g(n)$ satisfies a linear recurrence of order 2 with polynomial coefficients of degree 12, and offered a prize for a proof. We prove the recurrence. The proof proceeds by a deletion–insertion bijection reducing $g(n)$ to weighted enumerations of Kreweras-type lattice walks in the quarter plane, which we evaluate in closed form via the algebraic kernel method of Bousquet-Mélou and Mishna. Along the way we obtain several enumerative results of independent interest: a closed form for reverse-Kreweras walks ending on the diagonal, an explicit algebraic generating function for the diagonal, and the identity that the ballot-weighted sum $\sum_{a \geq c} \text{Bal}(a, c) k(3n - a - c; a, a - c)$ counts reverse-Kreweras walks of length $3n + 1$ ending at $(1, 0)$. The recurrence itself is explained structurally: g lies in a 2-dimensional module over $\mathbb{Q}(n)$ spanned by two hypergeometric terms, which forces a second-order recurrence and produces its coefficients by Cramer’s rule.

1 Introduction

A *solid standard Young tableau* (solid SYT) of a solid (three-dimensional) shape S with N cells is a bijective filling of the cells of S by $1, 2, \dots, N$ that increases in each of the three coordinate directions. Solid SYT were introduced and popularized by Zeilberger as a three-dimensional analogue of standard Young tableaux; in contrast to the classical case, no product formula is known, and the counting sequences appear to be substantially wilder.

We consider the two-layer shape $S_n = [[n, n], [n, 1]]$: the first layer consists of two rows of length n , the second layer of a row of length n atop the first row and a single cell atop the first cell of the second row; $|S_n| = 3n + 1$. Write $g(n)$ for the number of solid SYT of S_n . The sequence begins

2, 48, 1038, 22566, 500144, 11302300, 259808162, 6059911302, ...

*Purdue University; jpadhi@purdue.edu

Zeilberger computed $g(n)$ for $n \leq 40$ and observed empirically that the sequence is P -recursive of order 2 with coefficients of degree 12; his *First Rigorous Challenge* asks for a proof. The present paper provides one.

Theorem 1.1. *Let $K(n) = \frac{4^n(3n)!}{(n+1)!(2n+1)!}$ be the n -th Kreweras number, and set*

$$c(n) = \frac{8n^2 + n - 24}{4(n+2)(2n+3)}, \quad T(n) = \frac{7n+5}{5} \prod_{k=1}^n \frac{6(2k-1)(6k-1)(6k+1)}{(k+1)(4k+3)(4k+5)}.$$

Then for all $n \geq 1$,

$$g(n) = c(n)(3n+1)K(n) + T(n).$$

Consequently g lies in the 2-dimensional $\mathbb{Q}(n)$ -module spanned by the hypergeometric terms $(3n+1)K(n)$ and $T(n)$, and therefore satisfies a second-order linear recurrence with polynomial coefficients; the resulting operator, computed by Cramer's rule and displayed in Appendix A, has coefficients of degree 12 and coincides with Zeilberger's empirical recurrence. The recurrence has been verified against the first 56 terms of the sequence.

The route to Theorem 1.1 passes through the enumeration of *reverse Kreweras walks*: quarter-plane walks with steps $E = (1, 0)$, $N = (0, 1)$, $D = (-1, -1)$. Writing $k(m; i, j)$ for the number of such walks of length m from the origin to (i, j) , we prove two evaluation lemmas which we believe are of independent interest.

Theorem 1.2 (Diagonal evaluation). *For all $n \geq 0$,*

$$\sum_{a=0}^n k(3n-a; a, a) = T(n).$$

Moreover the individual diagonal counts admit the closed form

$$k(3n-a; a, a) = K(n) \cdot \frac{(a+1)!(2a+1)!n!(3n-a)!}{4^a (a!)^3 (n-a)!(3n)!}, \quad 0 \leq a \leq n,$$

and the diagonal generating function is algebraic:

$$Q_d(x; t) := \sum_{m, a \geq 0} k(m; a, a) x^a t^m = \frac{1}{\sqrt{1-xT^2}} \left(\frac{T}{2t} + \frac{\sqrt{1-xT^2} - 1 + \frac{1}{2}xT^2}{tx^2} \right), \quad (1)$$

where $T \equiv T(t)$ is the unique power series satisfying $T = t(2 + T^3)$.

Theorem 1.3 (Ballot-weighted evaluation). *Let $\text{Bal}(a, c) = \frac{a-c+1}{a+1} \binom{a+c}{c}$ denote the ballot numbers. Then for all $n \geq 0$,*

$$S(n) := \sum_{0 \leq c \leq a} \text{Bal}(a, c) k(3n-a-c; a, a-c) = k(3n+1; 1, 0) = \frac{3(9n+16)(3n+1)}{8(n+2)(2n+3)} K(n).$$

The first equality in Theorem 1.3 – the ballot-weighted sum counts plain walks to $(1, 0)$ – falls out of the kernel method in two lines once the correct substitution is found (Section 5); we do not know a direct bijection and pose finding one as an open problem.

Method and attribution. Our main tools are the functional-equation and algebraic-kernel techniques developed by Bousquet-Mélou for the Kreweras model [2] and applied to the reverse model by Mishna [3]. The extraction that produces (1) appears to be new, as do the closed forms in Theorems 1.2 and 1.3. All symbolic computations (resultants, an exact ODE derivation in a radical function field, and two rational-function identities) were carried out in exact arithmetic and are reproducible from the accompanying code.

2 From tableaux to walks

Order the rows of S_n as $(1, 1), (1, 2), (2, 1)$ (the three rows of length n) and $(2, 2)$ (the single cell). A solid SYT of S_n is equivalent to a growth sequence $\emptyset = P_0 \subset P_1 \subset \cdots \subset P_{3n+1} = S_n$ of solid subshapes, each step adding one cell. Recording after each step the number of filled cells $x_{11}, x_{12}, x_{21}, x_{22}$ in the four rows, the constraints of solid-shape containment are precisely

$$n \geq x_{11} \geq x_{12}, \quad x_{11} \geq x_{21}, \quad x_{12} \geq x_{22} \cdot (\text{first cell}), \quad x_{21} \geq x_{22}, \quad x_{22} \leq 1.$$

More precisely, the cell of row $(2, 2)$ sits above the first cell of row $(2, 1)$ and in front of the first cell of row $(1, 2)$; it may be added exactly when $x_{12} \geq 1$ and $x_{21} \geq 1$.

Lemma 2.1 (Deletion–insertion). *Solid SYT of S_n are in bijection with pairs (w, τ) where w is a lattice walk of length $3n$ from $(0, 0, 0)$ to (n, n, n) with unit steps in the coordinates (x_{11}, x_{12}, x_{21}) staying in the cone $x_{11} \geq x_{12}, x_{11} \geq x_{21}$, and $\tau \in \{0, 1, \dots, 3n\}$ is an index with $x_{12}(\tau) \geq 1$ and $x_{21}(\tau) \geq 1$.*

Proof. Given a solid SYT, delete the cell of row $(2, 2)$ and record $\tau = (\text{its label} - 1)$, the number of cells added before it. What remains is a growth sequence of the shape $[[n, n], [n]]$, i.e. a walk w as described; the insertion condition at time τ is $x_{12}(\tau) \geq 1, x_{21}(\tau) \geq 1$ as noted above. The inverse re-inserts the cell after step τ and shifts later labels by one. $\square$

In the difference coordinates $u = x_{11} - x_{12}, v = x_{11} - x_{21}$ the three step types 11, 12, 21 become $(1, 1), (-1, 0), (0, -1)$ and the cone becomes the quarter plane: w is a *Kreweras excursion* of length $3n$ (in the reversed orientation, an excursion as well), whence the count $K(n)$ of Kreweras [1]. Complementary counting over the $3n + 1$ slots, using the $x_{12} \leftrightarrow x_{21}$ symmetry, gives:

Lemma 2.2. $g(n) = (3n + 1)K(n) - 2S(n) + T(n)$, where

$$S(n) = \sum_{\tau} \#\{w : x_{12}(\tau) = 0\}, \quad T(n) = \sum_{\tau} \#\{w : x_{12}(\tau) = x_{21}(\tau) = 0\}.$$

Lemma 2.3 (Prefix factorization). *With $k(m; i, j)$ the reverse-Kreweras counts as in the introduction,*

$$T(n) = \sum_{a=0}^n k(3n - a; a, a), \quad S(n) = \sum_{0 \leq c \leq a} \text{Bal}(a, c) k(3n - a - c; a, a - c).$$

Proof. If $x_{12}(\tau) = x_{21}(\tau) = 0$ the prefix consists of $a := \tau$ steps of type 11 only, leaving the state $(u, v) = (a, a)$; the completion is an arbitrary walk from (a, a) to $(0, 0)$ with steps $(1, 1), (-1, 0), (0, -1)$ in the quarter plane, of length $3n - a$. Reversing time turns it into a

reverse-Kreweras walk from the origin to (a, a) , giving $k(3n - a; a, a)$. If only $x_{12}(\tau) = 0$, the prefix uses a steps of type 11 and c steps of type 21 with $x_{11} \geq x_{21}$ throughout: a ballot sequence, $\text{Bal}(a, c)$ choices, ending at $(u, v) = (a, a - c)$; the completion contributes $k(3n - a - c; a, a - c)$ likewise. $\square$

Theorem 1.1 thus reduces to Theorems 1.2 and 1.3 (proved in Sections 3–5) together with the module computation of Section 6.

3 The reverse Kreweras model and two extractions

Let $Q(x, y; t) = \sum_{m, i, j \geq 0} k(m; i, j) x^i y^j t^m$ be the complete generating function of reverse-Kreweras walks. Constructing walks step by step (a D -step may not be appended to a walk ending on either axis) yields

$$(xy - t(x^2y + xy^2 + 1)) Q(x, y) = xy - R(x) - R(y) + c, \quad (2)$$

where $R(x) := tQ(x, 0; t)$ and $c := tQ(0, 0; t)$; note $Q(x, 0) = Q(0, x)$ by the $x \leftrightarrow y$ symmetry, and the $+c$ arises from inclusion–exclusion at the origin. The kernel $K(x, y) = xy - t(x^2y + xy^2 + 1)$, viewed as a quadratic in y , has roots with

$$Y_0 Y_1 = \bar{x}, \quad Y_0 + Y_1 = \frac{1}{t} - x, \quad \Delta(x) := (1 - tx)^2 - 4t^2 \bar{x}$$

(the discriminant after normalization), $Y_0 = \frac{(1-tx) - \sqrt{\Delta(x)}}{2t}$ being the root that is a formal power series in t . The rational kernel $1 - t(x + y + \bar{x}\bar{y})$ is invariant under $(x, y) \mapsto (\bar{x}\bar{y}, y)$ and $(x, y) \mapsto (x, \bar{x}\bar{y})$, generating an orbit of six pairs, exactly as in [2, §2.3] and [3, §2.3].

Lemma 3.1 (Validity). *All manipulations of this section are identities of formal series in t whose coefficients are Laurent polynomials in x (and y). Indeed $[t^m]Q(x, y)$ is a polynomial of degree at most m in each variable, so the substitutions $(x, y) \mapsto (\bar{x}\bar{y}, y)$ and $(x, y) \mapsto (x, \bar{x}\bar{y})$ produce elements of $\mathbb{Q}[x, \bar{x}, y, \bar{y}][[t]]$ and the orbit combination is well defined. For the partial-fraction step, factor $K(x, y) = -tx(y - Y_0)(y - Y_1)$; the roots satisfy $Y_0 = t\bar{x} + O(t^2) \in t\mathbb{Q}[x, \bar{x}][[t]]$ and $1/Y_1 = xY_0 \in t\mathbb{Q}[x, \bar{x}][[t]]$ (from $Y_0 Y_1 = \bar{x}$), so the expansions $\frac{1}{y - Y_0} = \bar{y} \sum_{k \geq 0} (Y_0 \bar{y})^k$ and $\frac{1}{y - Y_1} = -\frac{1}{Y_1} \sum_{k \geq 0} (y/Y_1)^k$ converge coefficient-wise in t , each coefficient of t^m being a Laurent polynomial. Hence $[y^0]$ of any product appearing below is well defined and may be computed term by term; the same applies to the splittings $[\cdot]^{>0}$, $[\cdot]^{\leq 0}$ in x used in the proofs of Propositions 3.2 and 3.3.*

Substituting the pairs (x, y) , $(\bar{x}\bar{y}, y)$, $(x, \bar{x}\bar{y})$ into (2) (legitimate by Lemma 3.1), and forming the combination (first)–(second)+(third) so that only $R(x)$ survives among the unknown univariate series, we obtain

$$K_r \cdot [xyQ(x, y) - \bar{x}Q(\bar{x}\bar{y}, y) + \bar{y}Q(x, \bar{x}\bar{y})] = xy - \bar{x} + \bar{y} + c - 2R(x).$$

Dividing by the rational kernel K_r , expanding $1/K_r$ by partial fractions in y (valid in the annulus dictated by Y_0, Y_1), and extracting the constant term in y – only the middle term on the left contributes, producing the diagonal – yields the fundamental *diagonal equation*

$$-\bar{x} Q_d(\bar{x}; t) \sqrt{\Delta(x)} = c - \bar{x} - 2R(x) + 2xY_0. \quad (3)$$

Under $x \mapsto \bar{x}$ this is equation (2.4) of [3]; we verified (3) independently to order t^{11} .

Canonical factorization. The Laurent polynomial $x \Delta(x) = t^2 x^3 - 2tx^2 + x - 4t^2$ has one root $X_0 = 4t^2 + O(t^3)$ that is small, and two large roots; writing $\Delta = \Delta_0 \Delta_+(x) \Delta_-(\bar{x})$ in the frame of the Kreweras discriminant (substitute $x \mapsto \bar{x}$; see [2, §2.2]) one finds, with $T = t(2 + T^3)$ the series of Theorem 1.2,

$$\Delta_0 = \frac{4t^2}{T^2}, \quad \Delta_+(x) = 1 - xT^2, \quad \Delta_-(\bar{x}) = 1 - \bar{x}T\left(1 + \frac{T^3}{4}\right) + \bar{x}^2 \frac{T^2}{4}.$$

Two coefficient extractions from (3), after division by $\sqrt{\Delta_0 \Delta_-(\bar{x})}$, now separate the two unknowns. Extracting the *strictly positive* part in x eliminates every R -term (they carry only nonpositive powers) and solves for the diagonal:

Proposition 3.2. $Q_d(x; t)$ is given by the closed form (1). In particular $Q_d(0; t) = (4T - T^4)/(8t)$ recovers the excursion generating function.

Extracting the *nonpositive* part instead eliminates the diagonal and solves for the axis series:

Proposition 3.3.

$$R(\bar{x}) = \frac{1}{2} \left[c - x + \frac{\bar{x}}{t} - \bar{x}^2 + \left(x + T - \frac{2\bar{x}}{T} \right) \sqrt{\Delta_-(\bar{x})} \right].$$

Proof of Propositions 3.2 and 3.3. Both propositions follow from one splitting. Substituting $2xY_0 = x(1 - tx)/t - (x/t)\sqrt{\Delta(x)}$ into (3) and rearranging gives

$$\sqrt{\Delta(x)} \left(\frac{x}{t} - \bar{x} Q_d(\bar{x}) \right) = c - \bar{x} + \frac{x}{t} - x^2 - 2R(x), \quad (4)$$

and applying $x \mapsto \bar{x}$ (Lemma 3.1),

$$\sqrt{\Delta(\bar{x})} \left(\frac{\bar{x}}{t} - x Q_d(x) \right) = c - x + \frac{\bar{x}}{t} - \bar{x}^2 - 2R(\bar{x}). \quad (5)$$

The canonical factorization $\Delta(\bar{x}) = \Delta_0 \Delta_+(x) \Delta_-(\bar{x})$, with the three factors displayed above, is an identity of series in t (it is the splitting of the cubic $\bar{x} \Delta(\bar{x})$ according to the t -adic size of its roots, cf. [2, §2.2]; we also verified it directly by expanding both sides using $T = t(2 + T^3)$). Dividing (5) by $\sqrt{\Delta_0 \Delta_-(\bar{x})}$ yields the *split form*

$$\sqrt{\Delta_+(x)} \left(\frac{\bar{x}}{t} - x Q_d(x) \right) = \frac{c - x + \bar{x}/t - \bar{x}^2 - 2R(\bar{x})}{\sqrt{\Delta_0} \sqrt{\Delta_-(\bar{x})}}. \quad (6)$$

Now classify supports in x (all statements per coefficient of t^m). On the left, the unknown block $x Q_d(x) \sqrt{\Delta_+(x)}$ involves only powers $x^{\geq 1}$, since $Q_d(x) \in \mathbb{Q}[x][[t]]$ and $\sqrt{1 - xT^2} \in \mathbb{Q}[x][[t]]$. On the right, the unknown block $2R(\bar{x})/(\sqrt{\Delta_0} \sqrt{\Delta_-(\bar{x})})$ involves only powers $x^{\leq 0}$, since $R(\bar{x})$, $1/\sqrt{\Delta_-(\bar{x})} \in \mathbb{Q}[\bar{x}][[t]]$ and $1/\sqrt{\Delta_0} = T/(2t)$ is free of x .

Positive part. Applying $[\cdot]^{>0}$ to (6) kills the R -block:

$$x Q_d(x) \sqrt{\Delta_+(x)} = \left[\sqrt{\Delta_+(x)} \frac{\bar{x}}{t} \right]^{>0} - \left[\frac{T}{2t} \frac{c - x + \bar{x}/t - \bar{x}^2}{\sqrt{\Delta_-(\bar{x})}} \right]^{>0}.$$

Both extractions evaluate in closed form. Writing $\sqrt{\Delta_+(x)} = \sum_{j \geq 0} \binom{1/2}{j} (-xT^2)^j$, the first is $(\sqrt{1 - xT^2} - 1 + \frac{xT^2}{2})/(tx)$ (the terms $j \geq 2$). In the second, every product of $\{c, \bar{x}/t, \bar{x}^2\}$

with the $\bar{x}$ -series $1/\sqrt{\Delta_-(\bar{x})}$ has only powers $x^{\leq 0}$; the single term reaching a positive power is $(-x) \cdot [\bar{x}^0] \frac{1}{\sqrt{\Delta_-(\bar{x})}} = -x$, contributing $+xT/(2t)$ after the outer sign. Hence

$$x Q_d(x) \sqrt{\Delta_+(x)} = \frac{\sqrt{1-xT^2} - 1 + \frac{xT^2}{2}}{tx} + \frac{xT}{2t},$$

and dividing by $x\sqrt{\Delta_+(x)}$ gives exactly (1), proving Proposition 3.2.

Nonpositive part. Applying $[\cdot]^{\leq 0}$ to (6) instead kills the Q_d -block and determines R :

$$\frac{2R(\bar{x})}{\sqrt{\Delta_0}\sqrt{\Delta_-(\bar{x})}} = \left[\frac{T}{2t} \frac{c-x+\bar{x}/t-\bar{x}^2}{\sqrt{\Delta_-(\bar{x})}} \right]^{\leq 0} - \left[\sqrt{\Delta_+(x)} \frac{\bar{x}}{t} \right]^{\leq 0},$$

where now $[\sqrt{\Delta_+(x)}\bar{x}/t]^{\leq 0} = \bar{x}/t - T^2/(2t)$ (the terms $j \leq 1$) and $[-x/\sqrt{\Delta_-(\bar{x})}]^{\leq 0} = -x(1/\sqrt{\Delta_-(\bar{x})} - 1)$. Multiplying through by $\frac{1}{2}\sqrt{\Delta_0}\sqrt{\Delta_-(\bar{x})}$ and collecting terms yields the formula of Proposition 3.3. (We verified the collected form symbolically, and both propositions independently against the walk counts to orders t^{15} and t^{13} respectively; every displayed intermediate identity above was additionally machine-checked coefficient-wise.) Proposition 3.3 is equivalent to, and corrects a typographical corruption in the published form of, the axis result of [3]. $\square$

4 Proof of Theorem 1.2

Since a reverse-Kreweras walk to (a, a) has length $\equiv 2a \pmod{3}$, the substitution $x = t$ aligns exponents:

$$A(t) := Q_d(t; t) = \sum_{n \geq 0} \left(\sum_a k(3n - a; a, a) \right) t^{3n} = \sum_{n \geq 0} \tilde{T}(n) t^{3n},$$

where $\tilde{T}(n)$ denotes the left side of Theorem 1.2. By Proposition 3.2, A is an explicit element of the field $\mathbb{Q}(t, T, \sigma)$ with $\sigma = \sqrt{1-tT^2}$; eliminating σ and T by resultants gives the minimal polynomial $P(t, A)$ of degree 6 (Appendix B); P is a polynomial in t^3 , and $P(0, B) = 2B - 2$, so by Hensel's lemma A is the *unique* power-series root with constant term 1.

Differentiation is explicit in the tower: $T' = (2+T^3)/(1-3tT^2)$ and $\sigma' = -(T^2+2tTT')/(2\sigma)$, with reductions $T^3 = (T-2t)/t$ and $\sigma^2 = 1-tT^2$. Computing A', A'', A''' in the 6-dimensional basis $\{T^i \sigma^j\}$ and solving a linear system over $\mathbb{Q}(t)$ produces an *exact* inhomogeneous ODE

$$q_c(t) + q_0(t)A + q_1(t)A' + q_2(t)A'' + q_3(t)A''' = 0$$

with the polynomial coefficients of Appendix B. Every step of this derivation is exact rational-function arithmetic; as a safeguard the ODE was verified on the series of A to order t^{19} . Extracting coefficients converts the ODE into a recurrence supported on shifts $\{-6, -3, 0\}$ in the exponent of t , hence a second-order recurrence $\sum_{i=0}^2 \gamma_i(n) \tilde{T}(n-i) = 0$ valid for $n \geq 2$. Finally, one checks – an identity of rational functions, verified symbolically – that the hypergeometric ratio

$$\frac{T(m)}{T(m-1)} = \frac{6(2m-1)(6m-1)(6m+1)(7m+5)}{(m+1)(4m+3)(4m+5)(7m-2)}$$

of the claimed closed form satisfies the same recurrence, that the leading coefficient $\gamma_0(n)$ has no integer zeros $n \geq 2$, and that the initial values agree. By induction $\tilde{T}(n) = T(n)$ for all n . For

the per- a closed form, extract $[x^a]$ from (1): with $u = xT^2$, one has $[u^j](1-u)^{-1/2} = \binom{2j}{j}4^{-j}$ and

$$[u^j] \left(1 - \left(1 - \frac{u}{2} \right) (1-u)^{-1/2} \right) = - \binom{2j-2}{j-1} 4^{-(j-1)} \frac{j-1}{2j} \quad (j \geq 1),$$

an identity of binomial coefficients (checked symbolically). Hence

$$[x^a] Q_d = \frac{1}{t} \left[\frac{\binom{2a}{a}}{2 \cdot 4^a} T^{2a+1} - \frac{(a+1)\binom{2a+2}{a+1}}{2(a+2)4^{a+1}} T^{2a+4} \right],$$

and applying Lagrange inversion $[t^N]T^k = \frac{k}{N} \binom{N}{(N-k)/3} 2^{N-(N-k)/3}$ at $N = 3n - a + 1$ gives $k(3n - a; a, a)$ as an explicit two-term Gamma expression. That this expression equals the product form displayed in Theorem 1.2 – equivalently, that the ratio in a is $\frac{(a+2)(n-a)(2a+3)}{2(a+1)^2(3n-a)}$ – is an identity of Gamma quotients whose ratio to the claimed form simplifies to 1; we verified this simplification symbolically, and the values numerically for all $0 \leq a \leq n \leq 8$. $\square$

5 Proof of Theorem 1.3

Let $C(z) = 1 + zC(z)^2$ be the Catalan series. By Lambert's generalized binomial theorem, $\sum_{c \geq 0} \text{Bal}(b+c, c)z^c = C(z)^{b+1}$, and hence the ballot-weighted sum telescopes into a single substitution:

$$\sum_n S(n)t^{3n} = \text{CT}_x \left[C(t^2\bar{x}) Q(x, t\bar{x}C(t^2\bar{x}); t) \right],$$

where CT_x takes the coefficient of x^0 (all objects are Laurent series in x with polynomial coefficients per power of t , so this is well defined). Write $y^* = t\bar{x}C(t^2\bar{x}) = \frac{1-\sqrt{1-4t^2\bar{x}}}{2t}$. The series y^* contains only strictly negative powers of x , and satisfies

$$txy^{*2} - xy^* + t = 0, \quad \text{i.e.} \quad xy^* = t + txy^{*2}.$$

Substituting into the kernel gives the remarkable collapse

$$K(x, y^*) = xy^* - t(x^2y^* + xy^{*2} + 1) = -tx^2y^*.$$

Since $C(t^2\bar{x})/(ty^*) = x/t^2$, the functional equation (2) yields

$$C(t^2\bar{x}) Q(x, y^*) = -\frac{1}{t^2x} (xy^* - R(x) - R(y^*) + c).$$

Taking CT_x : the term xy^* has no x^1 -coefficient (as y^* has no constant term), $R(y^*)$ is a series in t and negative powers of x , and c is free of x ; only $R(x)$ survives:

$$\sum_n S(n)t^{3n} = \frac{1}{t^2} [x^1] R(x) = \frac{1}{t} \sum_m k(m; 1, 0) t^m. \quad (7)$$

This proves the first equality of Theorem 1.3: $S(n) = k(3n+1; 1, 0)$. For the closed form, extract $[x^1]$ from Proposition 3.3 (in the positive-variable frame): with $\beta = 1 + \frac{T^3}{4}$,

$$[x^1]R(x) = \frac{1}{2} \left[\frac{1}{t} - \frac{2}{T} - \frac{T^2\beta}{2} + \frac{T^2}{8} (1 - \beta^2) \right],$$

and under the rational uniformization $t = T/(2 + T^3)$ this equals

$$t^2 \cdot \frac{40t^2 - 18tT + (108t^3 - 1)T^2}{256t^5}$$

identically in T (a rational-function identity, verified symbolically). Thus $\sum_n S(n)t^{3n} = \frac{40t^2 - 18tT + (108t^3 - 1)T^2}{256t^5}$, and Lagrange inversion, $[t^N]T^k = \frac{k}{N} \binom{N}{(N-k)/3} 2^{N-(N-k)/3}$, gives

$$S(n) = \frac{1}{256} \left(-18 L(3n+4, 1) + 108 L(3n+2, 2) - L(3n+5, 2) \right) = \frac{3(9n+16)(3n+1)}{8(n+2)(2n+3)} K(n),$$

the last equality being a Gamma-quotient identity (verified symbolically; both sides are explicit products). $\square$

6 The module structure and the recurrence

By Lemma 2.2, Theorem 1.3 and Theorem 1.2,

$$g(n) = (3n+1)K(n) - 2S(n) + T(n) = c(n) (3n+1)K(n) + T(n), \quad c(n) = \frac{8n^2 + n - 24}{4(n+2)(2n+3)},$$

proving the identity of Theorem 1.1. Both $A(n) := (3n+1)K(n)$ and $T(n)$ are hypergeometric with explicit ratios

$$r_A(n) = \frac{6(3n-1)(3n+1)}{(n+1)(2n+1)}, \quad r_T(n) = \frac{6(2n-1)(6n-1)(6n+1)(7n+5)}{(n+1)(4n+3)(4n+5)(7n-2)},$$

and these ratios are not equal as rational functions, so the sequences A, T span a free $\mathbb{Q}(n)$ -module of rank 2 closed under the shift. The three vectors expressing $g(n), g(n+1), g(n+2)$ in the basis $(A(n), T(n))$,

$$w_i = \left(c(n+i) \prod_{j=1}^i r_A(n+j), \prod_{j=1}^i r_T(n+j) \right), \quad i = 0, 1, 2,$$

are forced to be linearly dependent; Cramer's rule $\gamma_i = (-1)^i \det(w_j, w_k)$ produces an explicit order-2 operator with polynomial coefficients, which after clearing denominators and content has degree 12 and is displayed in Appendix A. It agrees, up to a scalar, with the operator fitted from Zeilberger's data, and annihilates the computed sequence for all 56 known terms. This completes the proof of Theorem 1.1 and settles the First Rigorous Challenge. $\square$

7 Remarks and open questions

Remark 7.1. *The identity $S(n) = k(3n+1; 1, 0)$ equates a ballot-weighted double sum of general-endpoint walk counts with a single near-axis count. Our proof is a two-line kernel computation; a bijective proof would be desirable.*

Remark 7.2. *The closed form for diagonal-endpoint reverse-Kreweras walks (Theorem 1.2) complements Kreweras' classical axis formula and Bousquet-Mélou's diagonal result for the forward model [2, Thm. 2]; we have not found it in the literature. General endpoints provably admit no single hypergeometric closed form of comparable shape.*

Remark 7.3. *The Second Rigorous Challenge (non-holonomicity for the cylindrical shape $(2, 1, 1) \times [n]$) remains open; the walk model there lives in a union of three Weyl chambers of S_4 , outside the scope of reflection-based methods, and we hope to return to it.*

Acknowledgements. All symbolic and numerical verifications reported here are reproducible from the scripts at <https://github.com/jaideepsaipadhi/solid-syt-recurrence>. I thank Doron Zeilberger for posing the challenge.

References

- [1] G. Kreweras, Sur une classe de problèmes de dénombrement liés au treillis des partitions des entiers, *Cahiers du B.U.R.O.* **6** (1965), 5–105.
- [2] M. Bousquet-Mélou, Walks in the quarter plane: Kreweras’ algebraic model, *Ann. Appl. Probab.* **15** (2005), no. 2, 1451–1491.
- [3] M. Mishna, Classifying lattice walks restricted to the quarter plane, *J. Combin. Theory Ser. A* **116** (2009), no. 2, 460–477; arXiv:math/0611651.
- [4] S. B. Ekhad and D. Zeilberger, Computational and theoretical challenges on counting solid standard Young tableaux, arXiv:1202.6229 (2012); challenge statements and data on the accompanying webpage in the Personal Journal of S. B. Ekhad and D. Zeilberger.

A The order-2 operator

The recurrence is $p_0(m)g(m) + p_1(m)g(m-1) + p_2(m)g(m-2) = 0$ for $m \geq 3$, with:

$$p_0(m) = 258048m^{12} + 1169664m^{11} + 326816m^{10} - 5324160m^9 - 6298026m^8 + 6420627m^7 + 12567093m^6 - 6437$$

$$p_1(m) = -13934592m^{12} - 28325376m^{11} + 80053056m^{10} + 126497520m^9 - 162057066m^8 - 177863088m^7 + 13367$$

$$p_2(m) = 188116992m^{12} - 87899904m^{11} - 1694242656m^{10} + 1930644720m^9 + 3397410216m^8 - 5838018732m^7 -$$

B The exact ODE for the diagonal series

With $A(t) = Q_d(t; t)$: $q_c + q_0A + q_1A' + q_2A'' + q_3A''' = 0$, where:

$$q_c(t) = -81 (114t^3 - 5)$$

$$q_0(t) = 81 (3990t^6 + 282t^3 - 5)$$

$$q_1(t) = t (646380t^6 + 7845t^3 - 583)$$

$$q_2(t) = 24t^2 (10773t^6 - 99t^3 - 8)$$

$$q_3(t) = 16t^3 (3t - 1) (57t^3 + 1) (9t^2 + 3t + 1)$$